Link To Quiz: https://forms.gle/KAKpCNHgmF8cd3FK9

Quiz Question Rationale

Edge Detection Quiz

1. Zero-crossings in the second derivative

Correct: Second-derivative methods. Edges detected by zero-crossings are found where the second spatial derivative changes sign, which is the defining idea behind Laplacian-style detectors such as LoG.

Pitfalls: multi-stage detectors sounds plausible because Canny has several stages, but Canny is not defined by second-derivative sign changes; gradient-based methods and first-order methods are tempting because they also detect edges, but Sobel and Prewitt look for large first-derivative intensity changes instead of zero-crossings.

2. Prewitt compared with Sobel

Correct: Prewitt uses uniform weights across its 3x3 masks instead of Sobel's stronger centre row or column weights. This makes Prewitt a straightforward average-gradient operator.

Pitfalls: second-order derivatives describe LoG, not Prewitt; mandatory Gaussian smoothing belongs to Canny/LoG-style pipelines, not basic Prewitt; being more robust to noise than Sobel is a common reversal, since Sobel's centre weighting gives slight smoothing while differentiating.

3. Canny stage that thins broad responses

Correct: Non-maximum suppression. It compares each gradient response with neighbours along the gradient direction and keeps only local maxima, turning thick bands into narrow edge ridges.

Pitfalls: Gaussian smoothing is plausible because it changes edge appearance, but its purpose is noise reduction; hysteresis tracking improves contour continuity by linking weak and strong edges; double thresholding only labels pixels as strong, weak, or non-edge before hysteresis.

4. Notable disadvantage of LoG

Correct: It may produce double edges. Because LoG detects sign changes around rapid intensity transitions, a single physical boundary can create two nearby responses if smoothing scale or thresholding is not ideal.

Pitfalls: 'computationally simpler than Prewitt' is attractive because LoG can be implemented as one combined kernel, but it is still more complex than basic Prewitt; 'insensitive to noise' is wrong because second derivatives amplify fluctuations; 'cannot detect closed contours' is wrong because LoG can highlight contour structure.

5. Sobel edge direction formula

Correct: θ = tan^-1(Gy/Gx). The horizontal and vertical gradient responses form a gradient vector, so the inverse tangent of vertical over horizontal response gives its orientation.

Pitfalls: sin^-1(Gy/Gx) looks plausible because it uses a ratio, but sine is not the vector-angle relation used here; sqrt(Gx^2 + Gy^2) is gradient magnitude, not direction; Gx + Gy is only a crude combined response and has no geometric orientation meaning.

6. Purpose of Canny double thresholding

Correct: It classifies pixels as strong edges, weak candidate edges, or non-edges according to gradient magnitude. This prepares the map for hysteresis.

Pitfalls: smoothing before gradients is Gaussian smoothing; thinning wide edges to one-pixel ridges is non-maximum suppression; connecting weak edge pixels to strong ones is hysteresis tracking. These distractors are plausible because they are real Canny stages, but each names a different part of the pipeline.

7. Why Gaussian smoothing is used before LoG

Correct: It reduces noise that would otherwise be amplified by the second derivative. The Laplacian reacts strongly to small intensity fluctuations, so smoothing first prevents random texture from becoming false zero-crossings.

Pitfalls: computing edge orientation and strength is more natural for gradient methods; smoothing cannot guarantee that all zero-crossings form closed contours; it also does not make LoG behave like a first-order method, because the final detection still depends on the second derivative.

8. Practical advantage of Canny over Sobel and Prewitt

Correct: It produces thinner and more continuous edges. Non-maximum suppression thins responses, and hysteresis keeps weak edge pixels only when connected to strong edges, giving cleaner contours than single-mask operators.

Pitfalls: Canny does not avoid gradient information; it normally uses gradient magnitude and direction. It is not always faster because it has several stages, not a single convolution. It is not unique in detecting horizontal edges, since Sobel and Prewitt can detect them too.

9. Why Sobel is usually more noise-robust than Prewitt

Correct: Sobel assigns greater weight to the centre row and column, adding slight smoothing while differentiating. That extra weighting stabilises the response compared with Prewitt's uniform masks.

Pitfalls: both commonly use 3x3 kernels, so size is not the key difference; Sobel does not require an explicit Gaussian smoothing stage; relying on zero-crossings in the second derivative describes LoG, not a first-order Sobel operator.

10. When to prefer Prewitt

Correct: Use it when computational resources are limited and only a quick, rough edge map is needed. Prewitt is simple, light, and suitable for approximate detection on clean images.

Pitfalls: the most accurate localisation in very noisy images points toward Canny; essential zero-crossing detection points toward LoG; very thin and continuous edges regardless of computation time also points toward Canny, because Prewitt lacks non-maximum suppression and hysteresis.